NIGRAH Knowledge Graph & Prediction Engine (KGPE v1)

Core Hypothesis

Most fraud-detection systems are reactive.

They wait until:

```
Victims
↓
Complaints
↓
Investigation
↓
Warning
```

NIGRAH proposes:

```
Signals
↓
Patterns
↓
Graph Relationships
↓
Threat Emergence
↓
Early Warning
```

The system predicts risk from relationships, not just reports.

Why a Knowledge Graph?

Suppose we have three separate sources:

Source A

SEBI warning about fake advisors.

Source B

News article about Telegram investment groups.

Source C

Public website promising guaranteed returns.

Individually:

```
Low Intelligence
```

Together:

```
Potential Scam Ecosystem
```

The connection creates intelligence.

NIGRAH Graph Structure

The graph consists of nodes and relationships.

Node Types

Threat Signal Node

Examples:

```
Guaranteed Return
Urgency
Fake Authority
FOMO
Withdrawal Restriction
```

Manipulation Pattern Node

Examples:

```
Authority Exploitation
Greed Engineering
Trust Harvesting
Pressure Engineering
```

Scam Family Node

Examples:

```
Fake Trading App
Ponzi Scheme
Telegram Signal Fraud
Recovery Scam
```

Region Node

Examples:

```
Varanasi
Lucknow
Patna
Indore
```

Source Node

Examples:

```
SEBI
RBI
Court Order
News Report
Public Website
```

Platform Node

Examples:

```
Telegram
WhatsApp
Instagram
YouTube
Website
```

Example Graph

```
Guaranteed Return
        ↓
Greed Engineering
        ↓
Telegram Signal Fraud
        ↓
Uttar Pradesh
        ↓
Telegram
```

Now we can ask:

Which scam families are growing fastest in Uttar Pradesh?

Or:

Which manipulation patterns are increasingly appearing on Telegram?

The Prediction Engine

This is where things become interesting.

NIGRAH should not predict:

```
Scam / Not Scam
```

That is too simplistic.

Instead it predicts:

Threat Emergence Probability

Step 1

Signal Detection

Suppose we observe:

```
Guaranteed Return
```

```
Fake Authority
Urgency
VIP Access
Profit Screenshots
```

These become a signal vector.

Example:

```
[10,9,8,7,8]
```

Step 2

Pattern Construction

The system maps signals to manipulation patterns.

Result:

```
Authority Exploitation = High

Greed Engineering = High

Pressure Engineering = Medium

Social Validation = High
```

Step 3

Family Similarity

Now compare against known scam families.

Example:

Family	Similarity
Fake Advisor	91%
Telegram Signal Fraud	88%
Ponzi Variant	41%

Result:

```
Most Similar:
Fake Advisor Fraud
```

Step 4

Emergence Detection

Suppose:

Last month:

```
5 observations
```

Current month:

```
60 observations
```

Growth:

```
12x
```

Even if no regulator has warned about it yet:

NIGRAH flags:

```
Emerging Threat
```

The Emerging Threat Formula

This may become the signature innovation.

Emerging Threat Index (ETI)

```
ETI
```

```
=
```

```
Signal Novelty
×
Growth Velocity
×
Family Similarity
×
Geographic Spread
```

Normalized:

```
0–100
```

Example

New Fraud Type

Observed:

```
AI-generated advisor videos
+
Guaranteed returns
+
Telegram onboarding
+
Withdrawal restrictions
```

Similarity:

```
74%
```

Known families.

Growth:

```
320%
```

Geographic spread:

```
5 states
```

Result:

```
ETI = 92
```

Meaning:

Potential high-priority emerging threat.

Regional Intelligence Graph

One thing most hackathon teams will ignore:

Fraud behaves differently by geography.

Example:

```
UP:
Telegram scams

Maharashtra:
Fake trading apps

Bihar:
Ponzi-style referrals
```

The graph stores:

```
Threat
↓
Region
↓
Language
↓
Platform
```

This allows:

Localized Risk Intelligence

Which is especially valuable for Tier-2 and Tier-3 investors.

Language Layer

This is another opportunity.

Many scams use regional language persuasion.

Examples:

Hindi:

```
Guaranteed income
```

Bhojpuri:

```
Double your money
```

Tamil:

Different wording.

Same manipulation signal.

The graph should store:

```
Signal
↓
Language Variant
```

This makes the system multilingual without requiring separate scam databases.

The NIGRAH Intelligence Cycle

```
Public Intelligence
        ↓

Signal Extraction
        ↓

Pattern Mapping
        ↓

Family Classification
        ↓

Threat Scoring
```

```
        ↓

Knowledge Graph Update
        ↓

Emerging Threat Detection
        ↓

Early Warning
```

What Is Actually Innovative?

Most fraud projects do:

```
Screenshot
↓
AI
↓
Scam
```

NIGRAH does:

```
Intelligence Sources
↓
Threat Signals
↓
Manipulation Patterns
↓
Scam Families
↓
Knowledge Graph
↓
Threat Index
↓
Emerging Threat Prediction
```

The intelligence does not come from user reports.

The intelligence comes from structured public evidence.

That is a much stronger answer when judges ask:

"Where does your model get its knowledge?"